PROMPT LIBRARY

Let AI Write the Code

Let AI write the code. Every lab works without coding in the Trading Simulator; use these prompts when you want to see, run or customise the code behind a lab. The same prompts, with Copy buttons, are in Prompt_Library.html.

- Click 'Open in Claude' or 'Open in ChatGPT': the prompt is filled in for you. Press Enter.
- Colab prompts: open Google Colab (colab.research.google.com) > New notebook, paste the code the AI wrote into a cell and press Run. Nothing to install.
- Strategy Builder prompts: paste the code into the simulator's Strategy Builder, press Validate, then Save - it then appears in Backtest, Diagnostics and Paper Trading.
- If the code stops with an error, paste the error back into the same chat and ask the AI to fix it.
- Never paste API keys into an AI chat. Paper trading only - nothing here is investment advice.

Lab 01 - Market sessions and your first data

Pipeline Skeleton, Configuration and the Alpaca Paper Account

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: show the current US market session (pre-market, regular, after-hours, overnight or closed) in New York time and the matching Singapore time using zoneinfo; then download SPY daily bars for 2018-01-01 to 2025-12-31 and print the first and last 5 rows and the number of bars.

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Lab 02 - Validate prices: adjusted versus unadjusted

Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: download AAPL and D05.SI (DBS) daily bars for December 2025 twice - unadjusted (auto_adjust=False) and adjusted (auto_adjust=True). For 2025-12-31 print open, high, low and close from both, the ratio adjusted/unadjusted close (the dividend adjustment factor), and explain why TradingView's default chart matches the unadjusted prices.

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Lab 03 - Multi-timeframe data, a data-quality report and headline sentiment

Ingestion Stage: Multi-Timeframe Bars, Headlines and Data Quality

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: download SPY daily bars for 2018-2025 and hourly bars for the last 60 days; resample the daily bars to weekly and monthly; print a data-quality report (missing values, duplicate timestamps, gaps longer than 4 calendar days, daily moves above 15% that may be unadjusted splits). Then get recent NVDA news headlines with yfinance (Ticker('NVDA').news), score each headline with VADER sentiment (vaderSentiment) and print the average sentiment per day.

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Lab 04 - Test a trading hypothesis

Hypothesis Stage: from a Written Card to an Automated Statistical Test

Prompt
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Lab 05 - Indicators and their Pine Script twin

Indicator Stage: Pine Script v6 Indicators and their Python Twins

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: for SPY 2018-2025 compute SMA(50), SMA(200), EMA(20), Bollinger Bands (20, 2), RSI(14) and MACD (12, 26, 9) with pandas only, print the latest values, and plot the price with the overlays and RSI and MACD in panes below. Then write the same indicators as a TradingView Pine Script v6 indicator I can paste into the Pine Editor.

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Lab 06 - A crossover backtest to compare with TradingView's Strategy Tester

Strategy Prototype: TradingView Strategy Tester, Paper Trading from the Chart and Pipeline Reconciliation

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: backtest an SMA 50/200 crossover on SPY 2018-2025 (long when SMA50 > SMA200), trade at the next day's open with a 0.05% cost per trade. Print the list of trades (entry and exit dates and prices), the number of trades, net profit and maximum drawdown. Then write the same strategy in TradingView Pine Script v6 using strategy() so I can compare it in the Strategy Tester.

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Lab 07 - A library of four strategy families

Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: backtest four strategies on 2018-2025 daily data, trade at the next day's open with a 0.05% cost per trade: an EMA 20/50 crossover on SPY; Bollinger mean reversion on KO (buy below the lower band, exit at the middle band) with k = 2.0 and k = 1.5; 126-day time-series momentum on QQQ; and a Donchian 20/10 breakout on QQQ. Print CAGR, Sharpe ratio, maximum drawdown and number of trades in one table and explain when each family tends to work.

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Lab 08 - Place paper orders on Alpaca

Execution Stage: an Order Manager that Trades the Paper Account in Any Session

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: using alpaca-py with a PAPER account (TradingClient(..., paper=True)), read my keys from Google Colab Secrets named APCA_API_KEY_ID and APCA_API_SECRET_KEY (never put keys in the code), then place: a market order to buy 1 SPY; a limit order to buy 1 SPY 2% below the last price with extended_hours=True and time in force DAY; and a bracket order with a 3% take-profit and a 2% stop-loss. Print each order's id and status, list the open orders, then cancel the limit order.

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Lab 09 - Performance and risk metrics against buy-and-hold

Evaluation Stage: Performance and Risk Metrics with a Benchmark Table

Prompt
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Lab 10 - Grid search, out-of-sample ranking and decay

Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

Prompt
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Lab 11 - A walk-forward machine-learning signal

ML-Signal Stage: a Walk-Forward Classifier from Probability to Position

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: predict whether SPY closes higher tomorrow with walk-forward logistic regression and random forest classifiers (expanding window, first 504 days to train, retrain every 126 days), using features computed with data up to today only: RSI(14), 5- and 21-day returns, 21-day volatility and the distance from SMA(50). Report accuracy and AUC per model and the feature importance, and backtest 'long when P(up) > 0.55' (trade at the next day's open with a 0.05% cost per trade) against buy-and-hold.

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Lab 12 - Market regimes with PCA and K-means

Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: label SPY market regimes for 2018-2025 with PCA and K-means (3 clusters) on the 21-day return, 21-day volatility and 63-day return, fitting on past data only (refit once a year). Plot the regimes on the price chart, then backtest an EMA 20/50 crossover with and without a gate that stays flat in the high-volatility regime (trade at the next day's open with a 0.05% cost per trade) and compare CAGR, Sharpe and maximum drawdown.

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Lab 13 - A/B test two strategy variants

A/B-Test Stage: a Statistical Decision before Capital Moves

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: A/B test two strategies on SPY 2018-2025 (trade at the next day's open with a 0.05% cost per trade): A = EMA 10/30 crossover, B = EMA 20/50 crossover. Compare their daily returns with a Welch t-test, a paired t-test and a block-bootstrap 95% confidence interval of the Sharpe difference, apply a Bonferroni correction for 20 variants tried, and state the decision on H0 in one sentence.

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Lab 14 - A hedged pairs trade

Pairs Stage: a Hedged, Market-Neutral Statistical-Arbitrage Sleeve

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: screen KO/PEP, XOM/CVX, V/MA and HD/LOW for cointegration (Engle-Granger) on 2018-2022 and estimate each hedge ratio by OLS. Then backtest a market-neutral KO/PEP spread strategy on 2023-2025: 60-day z-score of the spread, enter when |z| > 2, exit when |z| < 0.5, long one leg and short the other, 0.05% cost per leg. Print the screen table, the trades, CAGR, Sharpe, maximum drawdown and the strategy's beta to SPY.

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Lab 15 - Trading costs and the break-even cost

Backtest-Engine Stage: Cost Models, Execution Timing and Break-Even

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: backtest 'buy when RSI(2) < 10, sell when RSI(2) > 70' on SPY 2018-2025 at the next day's open under three cost models - zero, retail (0.02% spread + 0.01% slippage) and high (0.10% + 0.05%). Print gross and net CAGR and Sharpe for each, find the break-even one-way cost in basis points by bisection, and compare next-day-open with next-day-close execution.

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Lab 16 - A risk-checked paper-trading bot

Automation Stage: a Scheduled, Risk-Checked Forward Test and Live Paper Trading

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: write a simple paper-trading bot for SPY: each run it takes daily bars up to yesterday, computes an EMA 10/30 crossover, checks risk limits (maximum position 50% of equity, no new buys after a 2% daily loss, a KILL_SWITCH flag) and changes the position only when the target changes, logging every decision to a CSV. Include a replay mode that simulates 2025 day by day with a $100,000 paper account (no broker needed) and compares the forward result with a backtest of the same rule.

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Lab 17 - A webhook that turns TradingView alerts into paper orders

Event-Driven Automation: TradingView Webhook to Signal, Risk and Execution

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: write a small FastAPI app with a /webhook endpoint that receives a TradingView alert JSON {secret, ticker, action, price}, rejects a wrong secret, validates the fields, confirms the signal with a 20-day SMA check, caps the position size, and logs the paper order it would send. Test it inside Colab with FastAPI's TestClient by posting 7 sample alerts (valid buy, valid sell, wrong secret, missing field, unknown ticker, duplicate, oversized) and print each result.

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Lab 18 - Diagnose a too-good-to-be-true backtest

Diagnostics Stage: Automated Bias Checks before Go-Live

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: recreate this biased SPY strategy on 2018-2025: signal = 1 when the 5-day rolling mean computed with center=True is rising; strategy return = signal x the same day's return; no costs. Print its CAGR and Sharpe, then fix it one problem at a time - a trailing window, next-day execution, a 0.05% cost - printing CAGR and Sharpe after each fix next to buy-and-hold, and explain each bias in plain English.

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Lab 18 - Fix the Lab 18 exhibit in the Strategy Builder

Diagnostics Stage: Automated Bias Checks before Go-Live

Prompt
Here is a strategy from my Trading Simulator's Strategy Builder that looks too good to be true. Find the look-ahead bias, explain it in plain English for a beginner, and give me the corrected code with the same class name and params, changing as little as possible. Use only data up to each bar: the engine trades on the next bar. Here is the code: [paste the exhibit's code from the Strategy Builder here]

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Lab 19 - A one-page performance tearsheet

Monitoring and Reporting Stage: Automated Tearsheet and Backtest-vs-Forward Report

Prompt
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: produce a one-page PDF tearsheet for an EMA 10/30 crossover on SPY 2018-2025 (trade at the next day's open with a 0.05% cost per trade): equity curve against buy-and-hold, a drawdown chart, a monthly returns heatmap and a metrics table (CAGR, Sharpe, Sortino, maximum drawdown, win rate). Then replay 2025 day by day with the same rule and explain any difference from the 2025 backtest.

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Lab 20 - Your own strategy for the Strategy Builder

Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval

Prompt
Write a strategy for my Trading Simulator's Strategy Builder. Rules: one Python class that subclasses Strategy (from trader.strategies.base import Strategy, enter_exit, flat_until_ready); import indicators with `from trader.core import indicators as ta` (ta.sma(series, window), ta.ema(series, span), ta.rsi(close, window), ta.macd(close, fast, slow, signal) returns columns macd/signal/hist, ta.bollinger(close, window, k) returns columns lower/middle/upper/pct_b, ta.atr(df, window), ta.donchian(df, window), ta.realized_vol(close, window)); give it a name and a params dict such as {"window": {"default": 20, "min": 2, "max": 200, "help": "Lookback in bars"}}; implement generate_signals(self, data) where data is an OHLCV DataFrame, and return a Series of 1 (long), 0 (flat) or -1 (short) aligned to data.index. Use only data up to each bar - no center=True and no shift(-1) - because the engine trades on the next bar. Reply with the code only, then 3 bullet points explaining the rule. My idea: buy SPY when RSI(14) falls below 30 while the close is above its 200-day SMA, and sell when RSI(14) rises above 55. (Replace this idea with your own.)

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